One-Way Door

A GITHUB COPILOT SKILL · 0:00–0:40

Every merge is a door. Some only open one way.

Before you merge, One-Way Door answers the question no pipeline asks:

> **“If this goes wrong after deploy, can we actually roll back?”**

⇄ **Two-way door: roll back any time**

→ **One-way door: no way back**



THE PROBLEM

Rollback is the safety net nobody tests

Code rolls back. **State doesn't.** Whatever the new version wrote, published or sent stays behind for the old version to live with.

Release ships
changes a table, an event or a format

›

New state is written
rows migrated, events published

›

Something breaks
the plan: roll back

›

Code goes back
the data stays

›

Old code, new state
won't start, or silently wrong

Databases

A column renamed, dropped or retyped; values converted in place.

Old code fails its schema check, or queries a column that's gone.

Events and messages

A field removed or renamed, or its units changed (pounds to pence).

Old readers see zero or null, and nothing errors.

Caches and stored state

A new serialisation format, new keys, rewritten idempotency records.

Old code can't read them: misses, or work done twice.

External contracts

Payments, webhooks or files sent to partners in a new shape.

Already received: a redeploy can't recall them.

Why it gets through

Every safeguard looks **forward**.

Tests	new code on a fresh database · **never old code on new data**
CI/CD	deploying forward · **not back after state changed**
Migrations	applied in order · **most have no undo**
Code review	the lines in the diff · **not readers outside it**
Monitoring	errors and alerts · **not wrong values that throw nothing**

THE SOLUTION

One folder that makes Copilot ask first

`.github/skills/one-way-door` loads when a PR touches a migration or an event, or someone asks “can we roll back?”.

1 Classify

Each change: two-way or one-way.

2 Time it

When the door closes, what breaks, the safe window.

3 Blast radius

Who else reads the data, even outside the PR.

4 Split it

Expand → migrate → contract → drop, generated on request.

Scripts check
offline, deterministic, JSON

+

Copilot judges
readers, old code vs new state

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A verdict
as a PR comment

GitHub Copilot reviewed PR #482 · used one-way-door

ONE-WAY Safe rollback window: **none**

1. Column renamed and converted in place: the old ledger **won't start**.
2. Events already on Kafka **can't be rolled back**.
3. A missing field becomes **£0.00, silently**.

Blast radius: reconciliation-batch **(not in this PR)**

Suggested split: 4 PRs, each reversible.

Adopt it in one step

```
cp -R .github/skills/one-way-door \
   your-repo/.github/skills/
```

Needs git and Node 20+. No servers, no accounts, no network calls.